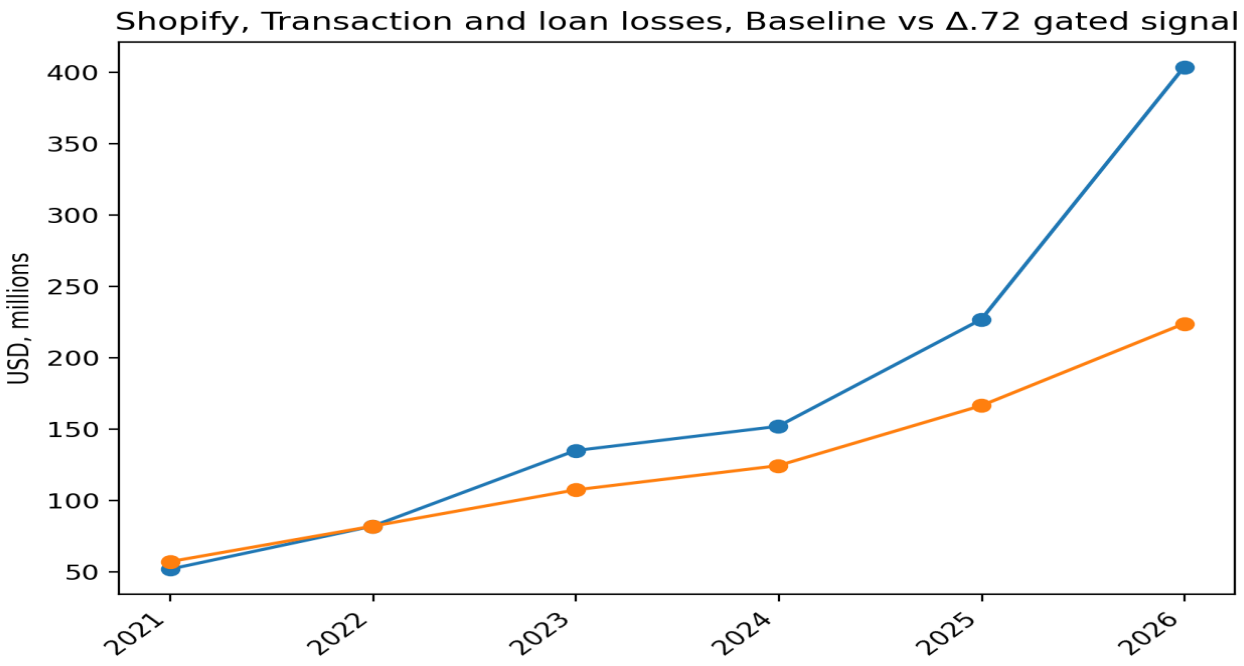
Δ.72 Coherence Engine, Open-Data Proof of Concept

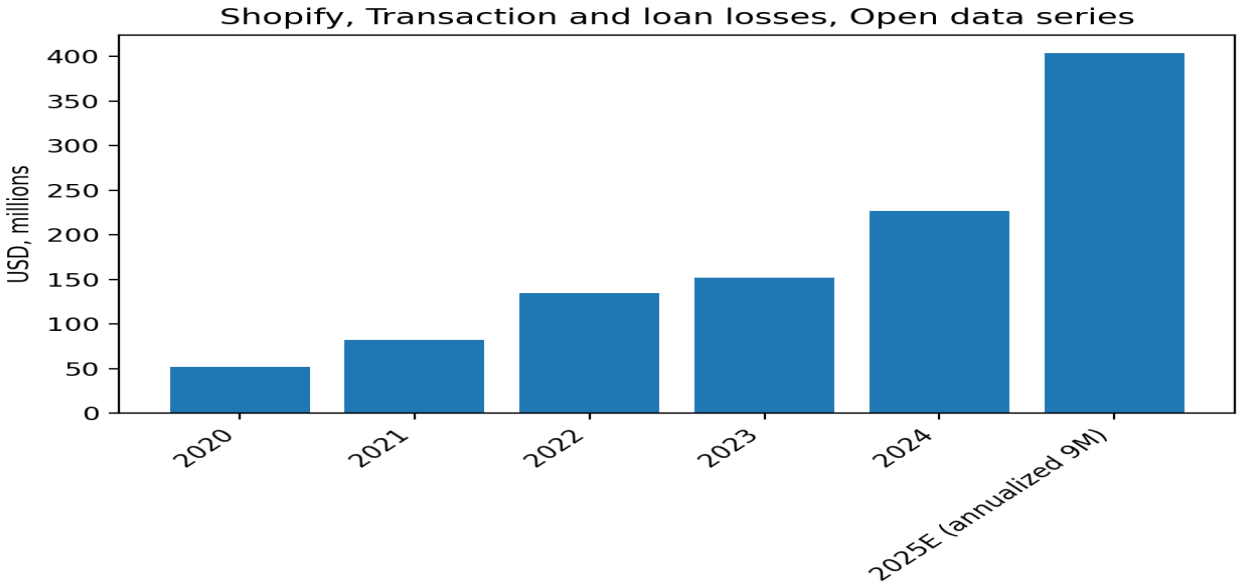
Company, Shopify (SHOP), KPI proxy for refunds, chargebacks, rework, Transaction and loan losses

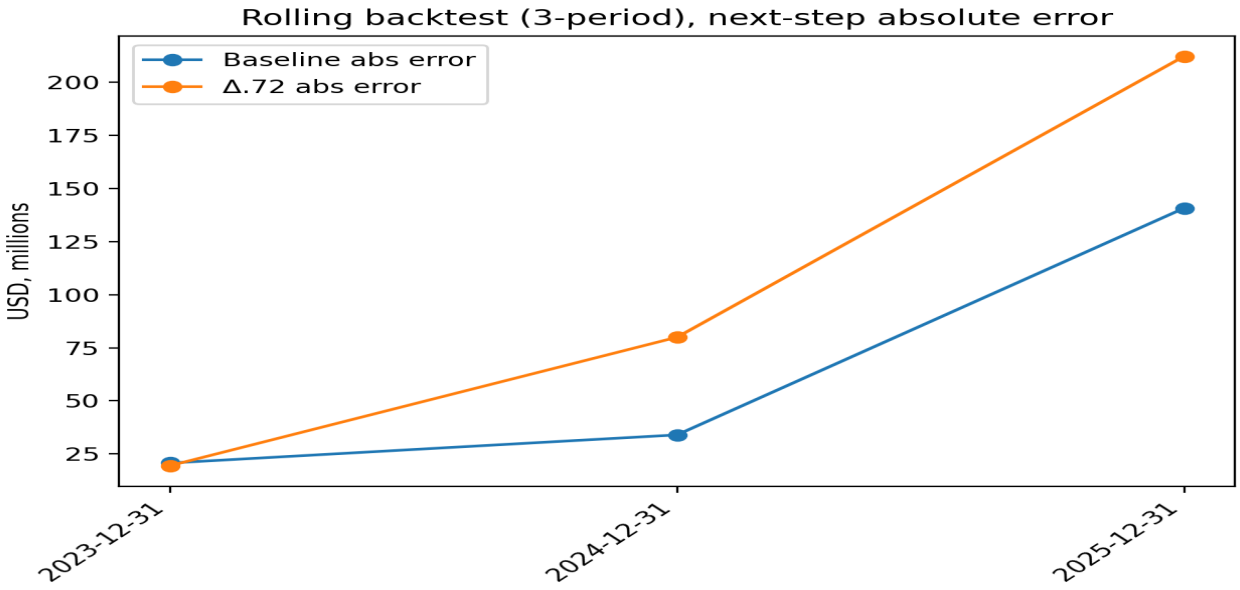
What this shows

We use Shopify's publicly filed transaction and loan losses as an open-data proxy for payment leakage, chargebacks, refunds, fraud, and rework-like loss. We apply a gated Δ.72 signal transform, score variance, THD proxy, recovery, and produce a Δ.72 summary block. Then we translate variance reduction into an estimated savings range using conservative capture rates.

Charts






Rolling backtest (3-period), next-step absolute error
Baseline abs error
Δ.72 abs error
USD, millions
200
175
150
125
100
75
50
25
2023-12-31
2024-12-31
2025-12-31

∆.72 Summary Block

metric	value
kpi_name	Shopify Transaction and loan losses (proxy for refunds, chargebacks, rework leakage)
baseline_mean_loss_usd_m	175.31
delta72_mean_loss_usd_m	126.92
efficiency_uplift_frac	0.2760
variance_reduction_frac	0.7747
recovery_time_baseline_steps	0.00
recovery_time_delta72_steps	0.00
thd_proxy_baseline	0.48901051689042774
thd_proxy_delta72	0.26392152659866946
loss_pool_basis_year	2024
loss_pool_usd_m	227.00
estimated_savings_usd_m_low_capture_0_10	17.59
estimated_savings_usd_m_mid_capture_0_25	43.97
estimated_savings_usd_m_high_capture_0_40	70.34

Savings translation (conservative)

capture_rate	basis_loss_pool_2024_usd_m	variance_reduction_frac_used	estimated_annual_savings_usd_m
10%	227.00	0.7747	17.59
25%	227.00	0.7747	43.97
40%	227.00	0.7747	70.34

Appendix A, Raw open-data series

period	transaction_and_loan_losses_usd_m	kind
2020	51.849	FY
2021	82.000	FY
2022	135.000	FY
2023	152.000	FY
2024	227.000	FY
2025-09-30 (9M)	303.000	9M
2025E (annualized 9M)	404.000	FY_estimate

Appendix B, Rolling backtest detail (window=3)

period	date	actual_usd_m	pred_baseline_usd_m	pred_delta72_usd_m	abs_err_baseline	abs_err_delta72	err_reduction_frac

2023	2023-12-31	152.00	172.77	132.49	20.77	19.51	0.0606
2024	2024-12-31	227.00	193.00	147.05	34.00	79.95	-1.3514
2025E (annualized 9M)	2025-12-31	404.00	263.33	191.89	140.67	212.11	-0.5079

Notes, 2025E is an annualization of 9M 2025, used only to keep the series continuous for illustration.